Semantic Differences from Apertus Hidden States

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Code, full technical report, and `make run`: github.com/amunozo/apertus-swissgov-rsd

Task

Given a parallel English page and its German, French, or Italian version from `admin.ch`, predict a score in [0, 1] for every whitespace token on both sides: how much its meaning differs from the other page. The metric is global token-level Spearman (×100) per language pair, macro-averaged over EN–DE, EN–FR, and EN–IT.

Method

We use **Apertus v1.5 8B** (`andreasmartin/apertus-v1.5-8b-text`, pinned revision) frozen, as a **reader** rather than a text generator. No text is generated in the submitted system.

1. **Four forward passes per document pair.** Each side X is read alone (`[BOS] NameX: X`) and conditioned on the other side Y (`[BOS] NameY: Y NameX: X`). X's subword ids are identical in both passes, so only the context changes.
2. **Middle-layer DiffAlign.** For each subword, $1 - \max_j \cos(x_i, y_j)$ against the other document's subwords, averaged per word, at layers 12–18 (raw and mean-centred, alone and conditioned).
3. **Contrastive surprisal.** $-\text{gain} = \log p(\text{word}) - \log p(\text{word} \mid Y)$: how little seeing the other document makes a word more predictable.
4. **Scalar head.** A logistic regression on 96 per-word features: the signals above, within-side moving averages over 1, 3, 7, and 15 words, document-level means, and 4 layout features. It is trained with soft-label cross-entropy and L2 ($\lambda = 10^{-3}$) on the 504 labelled dev documents. The 96 weights are the only trained parameters.

Engineering. Apertus runs in fp16 on Kaggle's free 2× T4 GPUs. Plain fp16 gave NaNs from layer 2 on: the quadratic xIELU activation and massive activations (about 119k) overflow fp16. We keep the residual stream and xIELU in fp32 and use exact per-token power-of-two scaling around `down_proj`. A CPU bf16 check gives per-layer cosine ≥ 0.9968. GPU features are cached as compact files (41.8 MB); `make run` refits the head and runs the official evaluation script in Docker on CPU in about a minute.

Results

The test split was used once, from the frozen git tag `final-run`. Baselines are the paper's released DiffAlign predictions, rescored with the official script on the same split (we rescored the seven full-document runs; the segmented LaBSE and XLM-R+SimCSE runs were not rescored).

System	Split	EN–DE	EN–FR	EN–IT	Mean
Ours: Apertus scalar head	test	**42.8**	**23.4**	**46.0**	**37.4** [31.9, 42.4]
DiffAlign Qwen3-Embedding-8B (best rescored)	test	19.7	11.2	20.4	17.1 [13.7, 20.5]
Ours: Apertus scalar head, nested 5-fold CV	dev	43.2	29.1	47.5	39.9
Ours: Apertus scalar head, dev/train → dev/val	dev/val	38.6	31.8	43.7	38.0
Apertus DiffAlign cond L14 + 7-word smoothing (no training)	dev	39.8	26.3	43.2	36.4
Apertus DiffAlign alone L14	dev	33.0	21.4	35.7	30.0
Qwen3-Embedding-8B DiffAlign, best layer (L21), our control	dev	30.6	19.7	33.0	27.8
Qwen3-Embedding-4B DiffAlign (best rescored on dev)	dev	20.6	11.7	20.2	17.5
Apertus DiffAlign last layer (L32, paper recipe)	dev	5.4	5.4	9.8	6.9
Apertus 3-shot JSON prompting (paper recipe)	dev (30 docs)	16.3	−10.2	3.6	3.2

What we found

- **Layer choice matters most.** The paper's recipe reads the last layer; for Apertus this gives 6.9. Layer 14 gives 30.0. A matched all-layer control with Qwen3-Embedding-8B shows the same effect inside Qwen (+11.3 [+10.1, +12.7] from L36 to L21). At each model's best layer and the same recipe, Apertus leads Qwen by only +2.2 [+1.8, +2.6] (+1.0 with smoothing). So most of the gain over the paper's baseline comes from the layer, not from the backbone.
- **Conditioning and smoothing help.** Conditioning on the other document adds +2.4 (30.0 → 32.4). A 7-word moving average adds +4.0, because gold differences come in contiguous spans; it helps Qwen similarly. Contrastive surprisal alone scores 19.6 (27.2 smoothed), while unconditioned $-\log p$ scores 0.0, so the signal is not word rarity.
- **Supervision adds a moderate gain.** The head adds +3.5 [+3.0, +3.9] in CV (every fold gains, +2.2 to +4.5) and +3.6 [+2.4, +5.0] on dev/val, mostly by sharper ranking within documents.
- **Reading beats generating.** With the paper's prompt, Apertus scores 3.2 on 30 short documents: 10 of 30 outputs are not valid JSON and the rest often have the wrong token count. Reading hidden states on the same documents scores 36.0–42.5, at about 1 s per forward pass instead of about 2.2 min per document.

Limitations

- The head is trained on dev, while the paper's baselines are training-free. Without any labels, our signal already scores 36.4 on dev vs. 17.5 for the best rescored baseline.
- Layers, windows, and head configuration were chosen on dev (only λ is nested), and we switched post hoc from per-pair to a shared head on a tie (40.0 vs. 39.9 CV; 37.6 vs. 38.0 dev/val). Dev CV is therefore slightly optimistic.
- EN–FR drops on test (23.4 vs. 29.1 in CV); we did not inspect test documents to diagnose it. Bootstrap CIs hold the fitted head fixed and understate the full uncertainty.
- The Qwen control covers unconditioned passes only; we did not run conditioned passes or a supervised head on Qwen.